A Pythagoras Tree Grown from a Dodecahedron

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Abstract

The Pythagoras tree of A. E. Bosman (1942) is drawn with squares in the plane and, in the three-dimensional versions we know of, with cubes. This article documents a tree built instead from regular dodecahedra: each dodecahedron carries five smaller copies, one on each edge of a single pentagonal "crown" face, scaled by $g \approx 0.4346$ and twisted by $45°$ about the edge. The construction first existed only as a pair of rendered images and two undocumented C++ programs. We recover its exact parameters from the original Blender scene, reformulate it as an iterated function system with condensation, and prove with a short computer-assisted argument that it never collides with itself: two of its dodecahedra overlap only when one is the parent of the other. A Pythagorean identity for the six face axes of the dodecahedron singles out the scale $1/\sqrt{5}$ and the twist $\arccos(1/\sqrt{5})$. With these values each child rests flat against a face of its parent, the five children together have exactly the parent's surface area, and the crown of the tree is a fractal of Hausdorff dimension exactly 2. Finally we ask whether the tree can grow on all twelve faces at once. A parity argument shows that at the artwork's proportions it cannot, but smaller scale factors admit certified clash-free all-faces trees.

Keywords: Pythagoras tree, dodecahedron, iterated function system, self-similar fractal, tight frame, mathematical art.
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1 Squares, cubes, and the missing dodecahedron

Bosman's tree starts with a square. On its top side sits a right triangle; on each leg of the triangle sits a new square, and the construction repeats on every new square. If the triangle has angle α, the two children are scaled by $\cos\alpha$ and $\sin\alpha$, and the theorem of Pythagoras says exactly that their areas add up to the area of the parent:

$$\cos^2\alpha + \sin^2\alpha = 1. \tag{1}$$

The tree owes its name to this identity [6]. It has been generalised to free angles and non-square shapes [2, 9, 10]. In three dimensions the natural analogue is built from cubes; the right triangle then becomes a tri-rectangular tetrahedron and (1) becomes de Gua's theorem on the squares of its face areas [3]. Cube trees appear as 3-D prints and even as model forests in turbulence simulations [11].

What does not seem to exist is a Pythagoras tree made of dodecahedra. We searched the scholarly indexes (arXiv full text, OpenAlex, Crossref, Semantic Scholar), the Bridges archive, code platforms and the open web and found none; every three-dimensional Pythagoras tree we located is built from cubes. The nearest relative is Whitehead's *generalised Pythagorean lute* [13], a string of dodecahedra scaled by the golden ratio, which is a lute rather than a tree. There is a good reason for the gap. The joint of a cube tree is a corner where three mutually

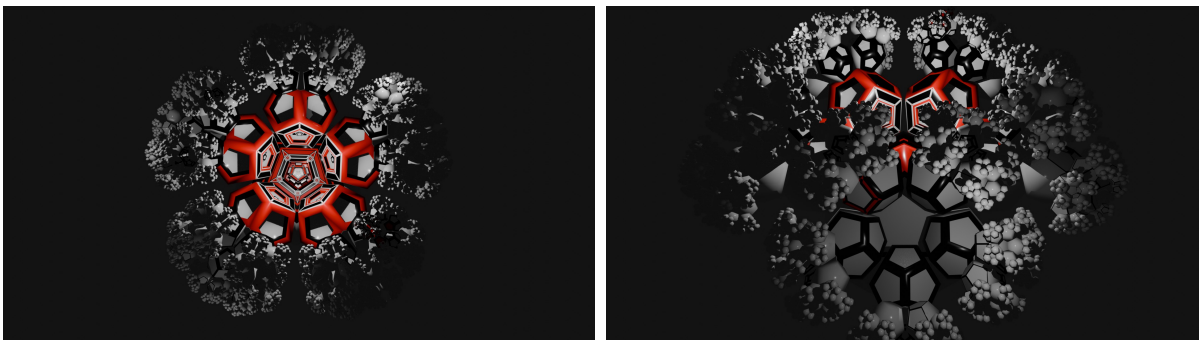


Figure 1: The artwork that started this article (rendered in Blender, March 2026). Left: looking down the axis of the crown face; right: the same tree from the side. The red frames are the first three generations, remodelled by hand; everything else is the unedited output of the program described in Section 2.

perpendicular planes meet, and any polygon with a corner at such a joint has a right angle there. Regular pentagons have 108° corners, so no joint built from regular dodecahedra can be right-angled, and the classical construction cannot be copied literally.

A dodecahedral tree must therefore decide what to keep from Bosman. The tree in Figure 1 keeps the recursion and the self-similarity and replaces the right-angled joint by a twist. This article is its documentation: how it is built (Section 2), how we recovered it from the files that produced the images (Section 3), why it does not collide with itself (Section 5), which member of the family deserves to be called Pythagorean (Section 6), and what happens when it is allowed to grow from every face (Section 7).

2 The construction

The base. Let $\varphi = (1+\sqrt{5})/2$. The twenty points

$$(\pm\varphi^{-1}, \pm\varphi^{-1}, \pm\varphi^{-1}), \quad (0, \pm 1, \pm\varphi^{-2}), \quad (\pm\varphi^{-2}, 0, \pm 1), \quad (\pm 1, \pm\varphi^{-2}, 0)$$

are the vertices of a regular dodecahedron B with edge $2\varphi^{-2} \approx 0.7639$, inradius $r_{\text{in}} \approx 0.8507$, circumradius $\sqrt{3}/\varphi \approx 1.0705$ and *midradius* 1: every edge midpoint is a unit vector. We fix one face F_0, the *crown*, with outward unit normal $a_0 = (0, 1, \varphi)/\sqrt{2+\varphi}$, and list its edges cyclically (counter-clockwise seen from outside) as $E_0, \dots, E_4$. For each k let m_k be the midpoint of E_k (a unit vector) and e_k the unit direction of E_k in the cyclic orientation.

The children. For a scale $g \in (0,1)$, an offset $d > 1$ and a twist angle θ, define five similarities

$$f_k(x) = g\ \mathrm{Rot}(e_k, \theta)\, x + d\, m_k, \qquad k = 0, \dots, 4, \tag{2}$$

where $\mathrm{Rot}(e, \theta)$ is the right-handed rotation by θ about the unit vector e. The child $f_k(B)$ is a copy of B shrunk by g, turned by θ about a line parallel to the edge E_k, and centred on the ray through the midpoint of E_k at distance d. The artwork uses

$$g = 0.36 \cdot \tfrac{1+\sqrt{2}}{2} \approx 0.43456, \qquad d = 1.125 \cdot \tfrac{1+\sqrt{2}}{2} \approx 1.35800, \qquad \theta = 45°. \tag{3}$$

(The factor $(1+\sqrt{2})/2$ is called `gold` in the original program; the golden ratio was probably intended, but these are the values that produced Figure 1.) The midpoint of E_k lies at distance $d - 1 \approx 0.358$ from the child's centre, which is less than the child's inradius $g\, r_{\text{in}} \approx 0.370$, so the parent's edge pokes into the child. This small overlap is the joint of the tree; it amounts to 0.16% of the child's volume. Figure 2 shows the first generations.

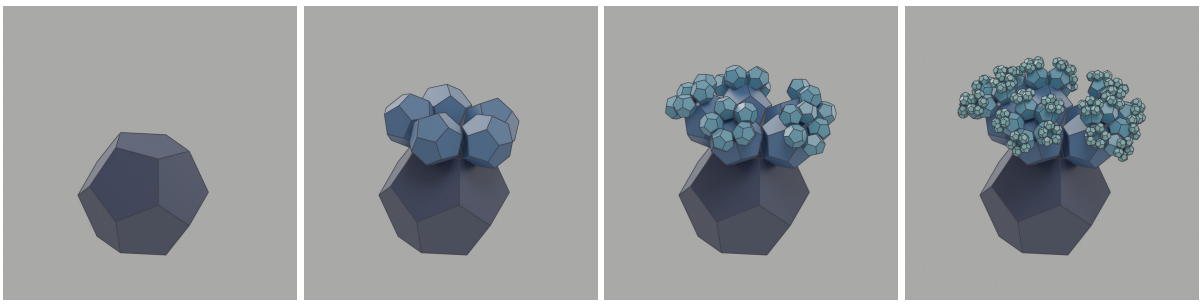


Figure 2: Generations 0–3 of the tree with the parameters (3): 1, 6, 31 and 156 dodecahedra. Each node sprouts five children on the edges of its own crown face.

The tree as a fixed point. Let $\mathcal{K}$ be the space of non-empty compact subsets of $\mathbb{R}^3$ with the Hausdorff metric. The map

$$W(X) = B \ \cup \ f_0(X) \cup \cdots \cup f_4(X)$$

is a contraction of $\mathcal{K}$ with ratio g, because each f_k is and the fixed set B does not increase Hausdorff distances. By Banach's fixed point theorem it has a unique fixed point, which we call the *dodecahedral Pythagoras tree*

$$T = B \cup f_0(T) \cup \cdots \cup f_4(T). \tag{4}$$

In Barnsley's terminology T is the attractor of an iterated function system *with condensation* B [1, 5]. Unwinding (4), T is the closure of the union of the *nodes* $f_w(B) = f_{w_1} \circ \cdots \circ f_{w_n}(B)$ over all finite words w in the alphabet $\{0, \ldots, 4\}$; the node $f_w(B)$ has depth n, scale g^n, and children $f_{wk}(B)$ attached to the edges of its own crown $f_w(F_0)$. The depth-n picture contains $(5^{n+1} - 1)/4$ dodecahedra; Figure 1 shows depth 5, that is, 3906 of them.

Remark 1. The original program recurses on absolute coordinates, so the node it reaches by choosing $k_1, \ldots, k_n$ is $f_{k_n} \circ \cdots \circ f_{k_1}(B)$, the composition taken in the opposite order. Both orders run over the same set of words, so they draw the same picture, but only the order used above makes a node's children sit on its own edges.

3 Recovering the artwork

The images in Figure 1 came with two C++ programs that write STL files, and the author no longer remembered which of the two had been used. We settled the question by checking each program against the actual geometry.

Both programs place the children as in (2), but they type the unit vectors m_k and e_k with two decimals (for instance $(0.5, 0.31, 0.81)$ for $(\frac{1}{2}, \frac{1}{2\varphi}, \frac{\varphi}{2})$), build the twist by conjugating a rotation about the z-axis with Euler rotations, and use a rotation-matrix routine that takes degrees and whose y-rotation is the transpose of the usual one. We ported the programs to Python with every one of these quirks and compiled the originals. On a solid five-child build of depth 3 (156 dodecahedra) the port and the compiled C++ agree to 5×10^{-7}, which is the print precision of the STL file. One quirk turned out to matter geometrically. The program writes $-45°$ for children 3 and 4, but their axes are typed with the opposite orientation, so all five twists agree: each is $+45°$ about the cyclically oriented edge, as in (2).

The Blender scene behind both images holds a single mesh of 148,164 polygons, 141,127 of them triangles. Splitting the triangles into connected components gives 7,909 pieces, of which 3,794 are clean regular dodecahedra (the rest are fragments and the polygons of the hand-built frames). Of these 3,794, 2,435 coincide with nodes of the depth-5 tree (4)–(3) to within 0.5% of their size. The other 1,359 are still exactly regular, but have been moved by

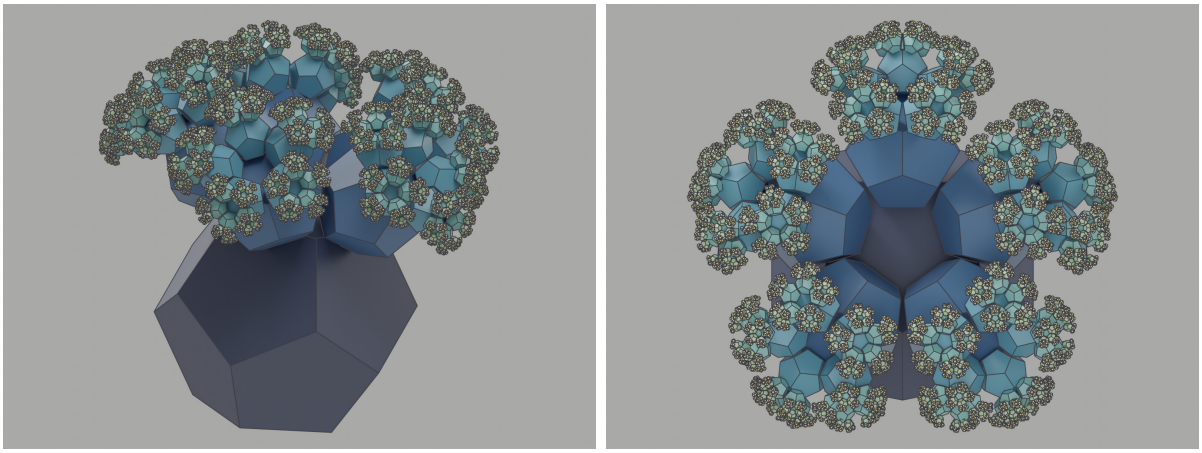


Figure 3: The artwork's tree regenerated from (2)–(3), depth 5, coloured by generation.

at most half their own size (median 5%) and turned by at most 1.1°, which we read as hand edits in Blender. Generations 0–2 were remodelled into the red frames. The second program, `DedocaedroA-copia.cpp`, passes radians where degrees are expected, so its twists are only 0.785°, and it matches none of the components. The artwork is therefore `DedocaedroTREEverCELLS.cpp` with all five children switched on, depth 5, and faces drawn solid. (As it stands, that file enables only children 0 and 1 and draws every edge as a thin strut, the "cells" of its name.) Replacing the typed two-decimal vectors by exact ones changes the maps by 0.4% and moves no vertex of the depth-5 tree by more than 0.007; from now on we use the exact version.

4 Symmetry and size

The rotations of B form the icosahedral group I of order 60; with the reflections it becomes the full group I_h of order 120. The tree keeps only the part of it that fixes the crown.

Proposition 2. *For every g, d, θ the tree T is invariant under the dihedral group D_5 of order 10 that fixes F_0: the rotations by multiples of 72° about a_0, and the five reflections in planes through a_0 and an edge midpoint m_k. For $\theta = 45°$ no other rotation of I preserves the family $\{f_k\}$.*

Proof. If σ is an isometry with $\sigma(B) = B$ and $\{\sigma f_k \sigma^{-1}\} = \{f_k\}$, then $\sigma(T)$ satisfies (4), so $\sigma(T) = T$ by uniqueness. A rotation ρ by 72° about a_0 sends (m_k, e_k) to (m_{k+1}, e_{k+1}), so $\rho f_k \rho^{-1} = f_{k+1}$. A reflection μ fixing F_0 reverses the cyclic orientation, so it sends e_k to $-e_j$ for the image edge E_j, and conjugation by a reflection reverses the sense of a rotation; the two reversals cancel: $\mu \,\mathrm{Rot}(e_k, \theta)\mu^{-1} = \mathrm{Rot}(-e_j, -\theta) = \mathrm{Rot}(e_j, \theta)$. The last claim we checked directly on the 60 elements of I. □

Generation n consists of 5^n dodecahedra of scale g^n. Counting volume and surface with multiplicity, the whole tree has

$$\frac{\mathrm{vol}(B)}{1 - 5g^3} \approx 1.696 \ \mathrm{vol}(B), \qquad \frac{\mathrm{area}(\partial B)}{1 - 5g^2} \approx 17.9 \ \mathrm{area}(\partial B)$$

for the artwork, and the surface series converges only because $5g^2 \approx 0.944 < 1$. The points that are limits of ever smaller nodes form the *crown set* K, the attractor of $\{f_0, \dots, f_4\}$ alone. Its similarity dimension s solves Moran's equation $5g^s = 1$, so $s = \log 5 / \log(1/g) \approx 1.931$. This equals the Hausdorff dimension of K as soon as the five pieces $f_k(K)$ are pairwise disjoint [8, 5, 4], and the certificate behind Theorem 3 below keeps them a positive distance apart.

5 The tree does not collide with itself

Children overlap their parents by design. We call any other pair of nodes whose interiors meet a *clash*. A tree is *self-avoiding* if it has no clashes at any depth.

Theorem 3. *With the parameters* (3)*, the dodecahedral Pythagoras tree is self-avoiding. Two of its dodecahedra have intersecting interiors if and only if one is the parent of the other. The same holds for the two-decimal version in the original program, and for its two-child variant.*

The theorem covers every depth, while a computer can only look at finitely many nodes. Self-similarity bridges the gap.

Lemma 4. *Let Q be a convex polytope with $B \subseteq Q$ and $f_k(Q) \subseteq Q$ for all k, and put $P = B \cup f_0(Q) \cup \cdots \cup f_4(Q)$. Suppose that*

(i) for $j \neq k$, every convex piece of $f_j(P)$ has interior disjoint from every convex piece of $f_k(P)$; and

(ii) $\operatorname{int} B \cap f_k f_j(Q) = \emptyset$ *for all j, k.*

Then T is self-avoiding.

Proof. Since W maps subsets of Q into Q, the fixed point satisfies $T \subseteq Q$, and then $T = W(T) \subseteq P$. Take distinct nodes $f_u(B)$ and $f_v(B)$ and strip their longest common prefix w; since f_w is a similarity, this does not change whether their interiors meet. If one remaining word is empty and the other has length 1, the nodes are parent and child. If one is empty and the other is $kj\cdots$, the second node lies in $f_k f_j(T) \subseteq f_k f_j(Q)$ and (ii) separates it from B. Otherwise the words begin with different letters $j \neq k$, the nodes lie in $f_j(T) \subseteq f_j(P)$ and $f_k(T) \subseteq f_k(P)$, and (i) separates them. □

Proof of Theorem 3 (computer-assisted). We fix about 1,000 unit directions u_i (the facet normals of the convex hull of the depth-4 tree, plus 400 random directions) and look for a polytope $Q = \{x : u_i \cdot x \le h_i\}$ that contains B and its own images. Starting from the hull of the depth-4 tree we iterate $h_i \leftarrow \max\bigl(h_B(u_i), \max_k [\, u_i \cdot t_k + g\, h_Q(R_k^{\top} u_i)\,]\bigr)$, where $f_k(x) = gR_kx + t_k$ and h_Q is the support function of the current Q, evaluated on its vertices. The iteration settles to machine precision in a few dozen steps. We then enlarge every h_i by 10^{-9} and verify $B \subseteq Q$ and $f_k(Q) \subseteq Q$ directly on the vertices; the resulting Q has 1380 vertices. Conditions (i) and (ii) are disjointness statements about pairs of convex polytopes. Each is decided by a linear program that finds the largest ball contained in both polytopes, where a negative radius $-r$ means that the two polytopes can each be pushed outwards by r before they touch. A negative optimum therefore certifies not only disjoint interiors but a positive distance. For (3) the smallest margin in (i) is 7.3×10^{-3} and in (ii) it is 0.30, both far above floating-point error (the tree has diameter about 3.3). For the two-decimal constants the margins are 5.8×10^{-3} and 0.30, and for the two-child variant 1.1×10^{-2} and 0.31. An independent check agrees: an exact separating-axis test on all pairs of the depth-5 tree (after a bounding-sphere filter) finds exactly the 3905 parent–child joints and nothing else. □

The twist, and its sense, are essential. With $\theta = 0°$ or $\theta = 90°$ (and the same g, d), adjacent children already overlap at depth 1, and by depth 5 there are 7615 and 3905 clashes respectively. Twisting the other way, $\theta = -45°$, gives 6740 clashes by depth 4. The first generation cannot tell θ from $\theta + 180°$, because an edge direction is a two-fold axis of B, but the later generations can. Figure 4 (top) maps the parameter plane: each point is checked to depth 4 and, when clean, put through the certificate. For scales $g \ge 0.2$, the range we scanned, self-avoiding crown trees exist only for twists between roughly $-85°$ and $140°$; the largest certified scale, $g = 0.4625$, occurs near $\theta = 65°$. The artwork sits close to the edge of the safe region, which at $\theta = 45°$

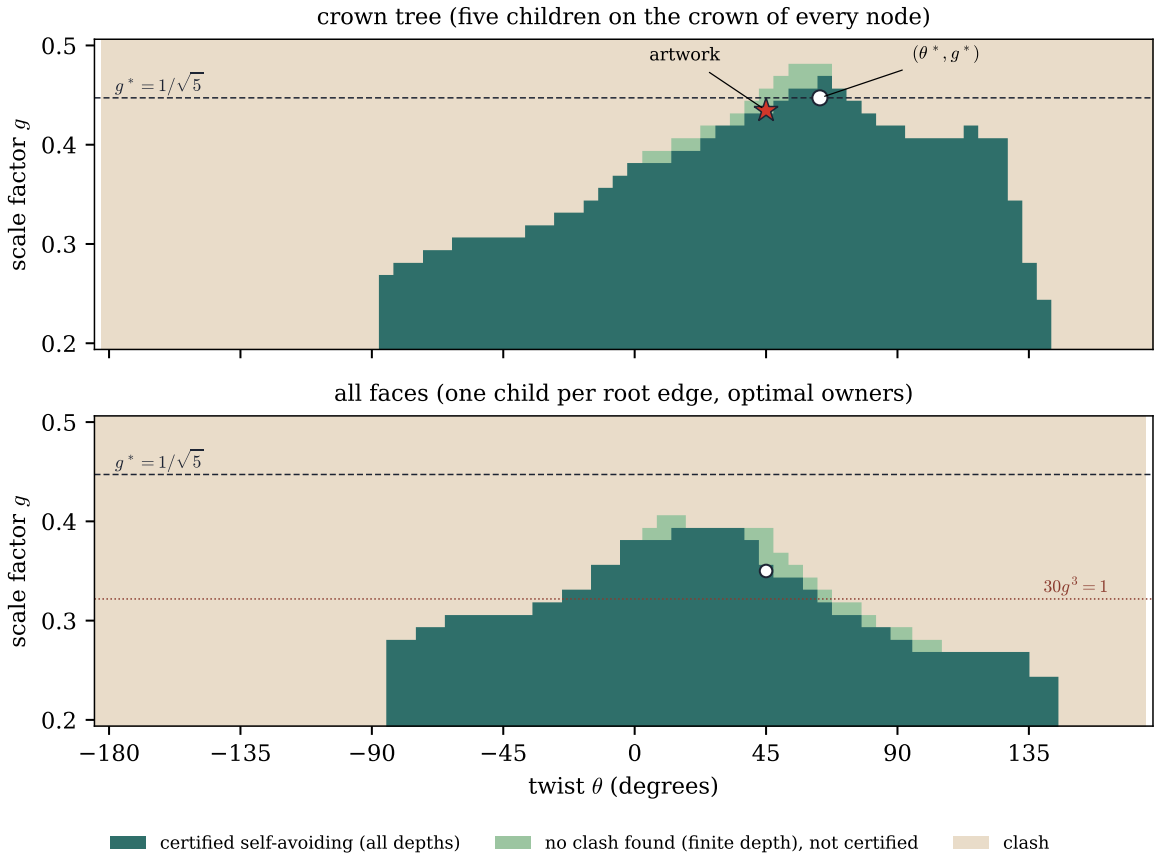


Figure 4: Where the tree is self-avoiding. Twist θ against scale g, with offset $d = 1 + \kappa g$, $\kappa = (d_{\mathrm{art}} - 1)/g_{\mathrm{art}} \approx 0.824$, so that every child keeps the artwork's joint. Top: the crown tree. Bottom: the best all-faces tree of Section 7. The dotted line in the bottom panel is the volume bound of Proposition 10 for trees that sprout on all thirty edges at every node. Twists of opposite sense give genuinely different trees, so the whole circle of twists is shown.

ends at a critical scale $g_c \in [0.4354, 0.4472]$: the scale 0.4354 is certified, and at 0.4472 a clash appears. The artwork's $g = 0.4346$ is certified as well. The upper value is $1/\sqrt{5}$, which will become important in the next section.

6 The Pythagorean dodecahedral tree

Which choice of g deserves the name Pythagoras? In Bosman's tree the scale of each child is the cosine of the angle between its growth direction and the parent's, and (1) says that these squared cosines sum to 1. The dodecahedron has its own version of (1). It is a classical fact about the six face axes, which form an equiangular tight frame [7, 12]; we include the short proof because the tree will use it.

Proposition 5. *Let $a_1, \dots, a_6$ be unit vectors along the six axes through opposite face centres of a regular dodecahedron. Then for every $u \in \mathbb{R}^3$*

$$\sum_{i=1}^{6} (u \cdot a_i)^2 = 2\,|u|^2. \tag{5}$$

In particular, if F_0 has normal a_0 and $a_1, \dots, a_5$ are the normals of the five faces adjacent to it, then $\sum_{k=1}^{5} (a_0 \cdot a_k)^2 = 1$ and each term equals $1/5$.

Proof. The matrix $S = \sum_i a_i a_i^\top$ is symmetric. It commutes with every rotation in I, because I permutes the six axes and the sign of a_i does not affect $a_i a_i^\top$. Each eigenspace of S is therefore invariant under I. But I fixes no line in $\mathbb{R}^3$: a line kept by a rotation of order five must be its axis, and I contains five-fold rotations about different axes. Hence it fixes no plane either (the normal of an invariant plane would be an invariant line). So S has a single eigenspace, $S = \lambda \,\mathrm{Id}$, and $3\lambda = \operatorname{tr} S = \sum |a_i|^2 = 6$. For the second statement, the six axes are equiangular, $|a_i \cdot a_j| = 1/\sqrt{5}$ for $i \neq j$, and $u = a_0$ gives $1 + 5 \cdot \frac{1}{5} = 2$. □

For the cube the three face axes form an orthonormal frame and (5) with constant 1 is the theorem of Pythagoras in coordinates; de Gua's theorem is its statement about areas [3]. Proposition 5 has the same reading: a planar region of area A casts shadows on the six face planes of the dodecahedron whose squared areas add up to $2A^2$. For the crown face itself this is a dodecahedral analogue of de Gua's theorem: the shadows of the five neighbouring faces on the plane of F_0 have area $A/\sqrt{5}$ each, and their squares add up to A^2.

If the five children of a node grow along the five neighbouring face axes and are scaled by the cosine of their tilt, which is the cosine rule of Bosman's tree with the five axes of Proposition 5 in place of two perpendicular legs, then their surface areas add up exactly to the parent's. It turns out that the twist of (2) can produce exactly this.

Proposition 6. *Let $g^* = 1/\sqrt{5}$, $\theta^* = \arccos(1/\sqrt{5}) \approx 63.43°$ (the angle between adjacent face axes) and $d^* = 1 + g^*$. For each k let F_k' be the face of B that meets the crown F_0 along E_k, with normal n_k. Then the child $f_k(B)$*

1. *has its crown axis along n_k, so it grows along a neighbouring face axis with scale $g^* = a_0 \cdot n_k$;*
2. *lies on the far side of the plane of F_k', its base face lying in that plane; the base face covers a small triangle of F_k' next to the edge E_k, so child and parent touch along that triangle but their interiors are disjoint.*

Proof. The vectors a_0, n_k and m_k are all orthogonal to e_k, the angle from a_0 to n_k is θ^*, and m_k bisects it. So $R = \mathrm{Rot}(e_k, \theta^*)$ maps a_0 to n_k, which is (1), and maps the normal $-a_0$ of the base face (the face opposite the crown) to $-n_k$. The base face of the child therefore lies in the plane $\{x \cdot n_k = c\}$ with $c = d^*\, m_k \cdot n_k - g^*\, r_{\mathrm{in}} = (1 + g^*) r_{\mathrm{in}} - g^* r_{\mathrm{in}} = r_{\mathrm{in}}$, using $m_k \cdot n_k = r_{\mathrm{in}}$ (the midpoint of E_k lies on F_k'). That plane is the plane of F_k', which supports B, so the two solids lie on opposite sides of it. In that plane the centre of the child's base face lies $g^* \rho$ beyond the line of E_k, where ρ is the inradius of a face, and the base face has a vertex pointing straight back at E_k. Since the circumradius R_f of a face exceeds ρ, that vertex crosses the edge line by $g^*(R_f - \rho) \approx 0.0555$, while the next two vertices stay outside. The contact region is therefore a triangle of area ≈ 0.0042, about 2% of the child's face. □

Moving the child any closer pushes the contact triangle into the parent, so $d^* = 1 + g^*$ is the *kissing distance*: the smallest offset at which child and parent do not overlap. (For convex bodies translated along a ray the offsets that produce an overlap form an interval, so this distance is well defined.) The kissing distance is exactly $1 + g$ for every g at $\theta \in \{0°, \theta^*, 90°\}$, and by the mirror symmetry of Proposition 2 also at $180° - \theta$. At $0°$ and $90°$ the reason is different from the face contact at θ^*: the child then presents an edge midpoint to the parent's edge midpoint (a parallel edge at $0°$, a crossed one at $90°$), and the plane $\{x \cdot m_k = 1\}$ separates the two solids once $d \geq 1 + g$. At other twists the ratio $(d - 1)/g$ at kissing varies: numerically it is 0.869, 0.874 and 1.015 at $20°$, $45°$ and $80°$. The artwork's 0.824 is below the kissing value at $45°$, which is the joint.

Theorem 7. *The tree T^* defined by (g^*, d^*, θ^*) has the following properties.*

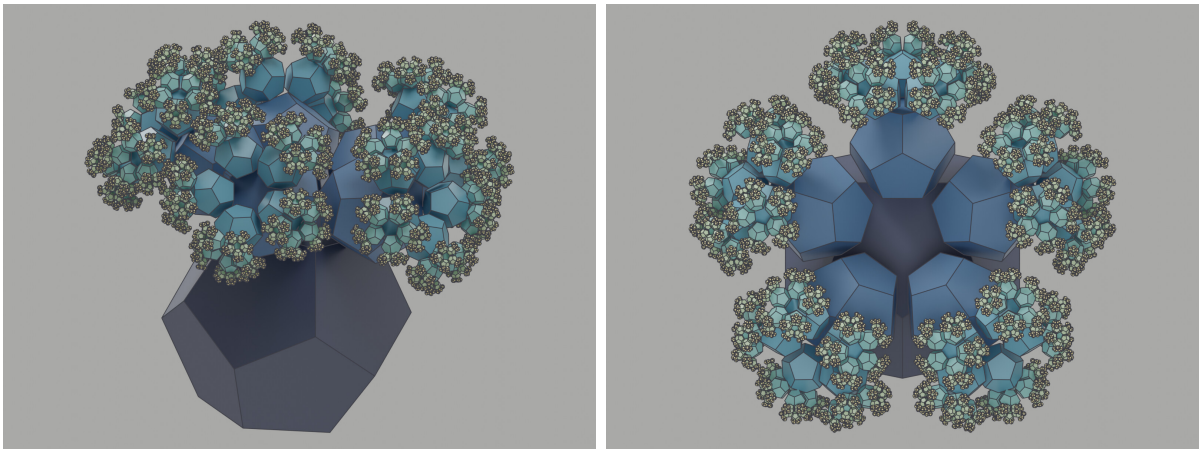


Figure 5: The Pythagorean dodecahedral tree T^* ($g = 1/\sqrt{5}$, $\theta = \arccos(1/\sqrt{5})$, $d = 1 + 1/\sqrt{5}$), depth 5. Every child stands on the plane of a lateral face of its parent and grows along that face's axis.

1. (Pythagorean) *The five children of every node have, together, exactly the surface area of their parent. Every generation has the same total surface area, and the total volume is* $\mathrm{vol}(B)/(1 - 5^{-1/2}) \approx 1.81\ \mathrm{vol}(B)$.

2. (Packing) *Its dodecahedra have pairwise disjoint interiors. Each child touches its parent along a triangle and touches nothing else.*

3. (Dimension two) *Its crown set K^* has Hausdorff dimension exactly* 2 *and finite, positive two-dimensional Hausdorff measure.*

Proof. (1) is $5(g^*)^2 = 1$. For (2), parent and child are separated by Proposition 6, and all other pairs by the certificate of Lemma 4, with margins 4.5×10^{-2} in (i) and 0.34 in (ii). For (3), $K^* \subseteq Q$ because Q is invariant under the maps, so $f_j(K^*) \subseteq f_j(P)$, and the positive margins in (i) show that the compact sets $f_j(K^*)$ are pairwise disjoint. This is the strong separation condition, and the dimension is the solution of $5\,(1/\sqrt{5})^s = 1$, that is, $s = 2$ [5, 4]. □

So the dodecahedron supports a Pythagoras tree after all. The right-angled joint is replaced by five equiangular directions, and $\cos^2\alpha + \sin^2\alpha = 1$ by $5 \cdot (1/\sqrt{5})^2 = 1$. Figure 5 shows T^*. With the artwork's joint offset $d = 1 + \kappa g^*$ in place of d^*, the scale g^* is certified self-avoiding at every twist we tested between 51.5° and 73° (in steps of 0.5°), and the certificate fails just outside this window. At the artwork's 45°, however, the ratio $1/\sqrt{5}$ produces clashes; the artwork's g lies 2.8% below it, on the safe side of g_c.

7 Growing from every face

The artwork grows from one face only. Could the tree grow from all twelve? The answer depends on what "growing from every face" means; we consider three readings.

Twelve crowns. Put a copy of the crown tree on every face, $\bigcup_j \gamma_j(T)$ with $\gamma_j \in I$ mapping F_0 to F_j (by Proposition 2 the choice of γ_j does not matter). Every edge lies on two faces, so it now carries two *twin* children with the same centre $d\,m$, twisted by $+\theta$ and $-\theta$ about the edge. The twins coincide when $\theta \in \{0°, 90°\}$ modulo 180°, because $\mathrm{Rot}(e, 2\theta)$ is then a symmetry of B, and cross each other otherwise; in both cases the tree clashes at depth 1. For the artwork's parameters there are 90 clashing pairs at depth 1 and 6390 by depth 4.

One child per edge. The natural repair keeps a single child on each of the 30 edges. This forces a choice of *owner* for every edge, one of its two faces. The child is twisted by $+\theta$ about the edge oriented counter-clockwise around its owner, and it carries the owner's subtree. Call two edges that meet at a vertex of a face F a *positive* pair of F if F owns both, and a *negative* pair if F owns neither. A rotation taking F to F_0 shows that the two children of a positive pair are congruent to two adjacent crown siblings with twist θ, and those of a negative pair to siblings with twist $-\theta$, equivalently $180^\circ - \theta$.

Theorem 8. *For every choice of owners there are at least six positive and at least six negative pairs. Both bounds are attained at once: there is an ownership with exactly six pairs of each kind.*

Proof. Let face F_j own o_j of its five edges. Each edge has one owner, so $\sum_j o_j = 30$ and $\sum_j (5 - o_j) = 30$. On the 5-cycle of edges of a face, c marked edges always include at least $\max(0, c-2)$ adjacent pairs, since at most two of five edges can be pairwise non-adjacent. Hence there are at least $\sum_j (o_j - 2) = 30 - 24 = 6$ positive pairs, and likewise $\sum_j (5 - o_j - 2) = 6$ negative ones. An integer program over the 2^{30} ownership choices finds an assignment with exactly six negative pairs (Figure 6, right); every face owns two or three edges, and counting shows that it has exactly six positive pairs as well. □

So any one-child-per-edge tree contains siblings twisted both ways, by θ and by $180^\circ - \theta$. For the artwork's g and d, a pair of adjacent siblings twisted coherently by θ is clash-free only for $\theta \in [38.3^\circ, 78.3^\circ]$ modulo 180° (Figure 6, left; computed on a 0.1° grid), and this interval contains no θ together with $180^\circ - \theta$: the larger of the two overlaps never drops below 0.058 of the child's size.

Corollary 9. *At the artwork's proportions, no one-child-per-edge extension to all twelve faces is clash-free, whatever the twist and the owners. At $\theta = 45^\circ$, optimal owners still leave exactly six clashing pairs at depth 1, and deeper generations add more (48 at depth 3).*

Smaller children escape the obstruction. Figure 4 (bottom) shows the certified region for the best owners. At $\theta = 45^\circ$ every scale we tested up to $g = 0.35$ (in steps of 0.0125) is certified self-avoiding at all depths, and the largest certified scale is $g = 0.3875$, at $\theta = 15^\circ$–35°. Figure 7 shows both sides of the boundary.

Every node sprouts everywhere. The most literal reading lets every node grow a child on each of its own 30 edges. Here a simple volume count rules out the artwork's scale.

Proposition 10. *If a tree with N maps of ratio g is self-avoiding, then $Ng^3 \le 1$.*

Proof. Nodes of the same depth are never parent and child, so the N^n nodes of depth n have disjoint interiors. They all lie in a bounded invariant set Q (a large enough ball will do), so $N^n g^{3n} \operatorname{vol}(B) \le \operatorname{vol}(Q)$ for every n, which forces $Ng^3 \le 1$. □

For $N = 30$ this requires $g \le 30^{-1/3} \approx 0.322$, whereas the artwork has $30g^3 \approx 2.46$. In fact every scale we tried, down to $g = 0.2$, clashes by depth 2, and in each case some grandchildren grow back into their grandparent.

8 Closing remarks

The artwork reached its proportions by eye, through a factor 0.36, a 45° twist and a constant called `gold` that is not the golden ratio. It sits just inside the region where the tree avoids itself, and 2.8% below the scale at which its surface area would be conserved from one generation to the next. Moving to that scale requires a larger twist, $\theta^* = \arccos(1/\sqrt{5})$, and then everything locks

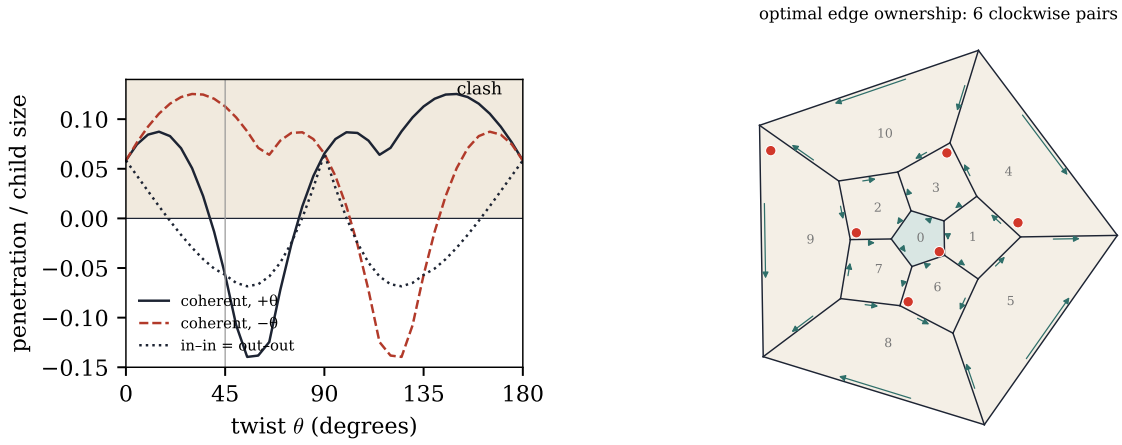


Figure 6: Left: overlap of two adjacent children (artwork's g, d) for the four relative orientations of their twists; values above zero are clashes. The two "coherent" curves are translates of each other by 180°, and no twist keeps both below zero. Right: an optimal edge ownership on a Schlegel diagram (face 11, opposite the crown, is outermost). Arrows show the counter-clockwise orientation around the owner, and dots mark the six unavoidable negative pairs.

into place: children rest flat on their parents' faces, the surface identity is the dodecahedron's own Pythagorean theorem, and the crown becomes a set of dimension exactly two.

Some questions remain open. The certificate decides individual parameter points; the exact shape of the boundary curves in Figure 4 is unknown, as is whether the kissing distance $1 + g$ occurs only at the twists we found. Non-uniform twists, unequal scales, or children on vertices instead of edges would all give new trees. The same constructions could be tried on the 120-cell, whose cells are dodecahedra.

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Code and data

All code and data are public at https://github.com/MiguelPerezI/dodecahedron-pythagoras-tree: the Python package `dodecatree` (geometry, the faithful port of the original programs, the exact construction, collision tests and the certificate), the original C++ sources, and scripts that rebuild the original programs, compare them with the Blender scene, run the clash study, and generate every figure and number in this article. STL and OBJ models of all trees shown can be regenerated with `scripts/make_tree.py`.

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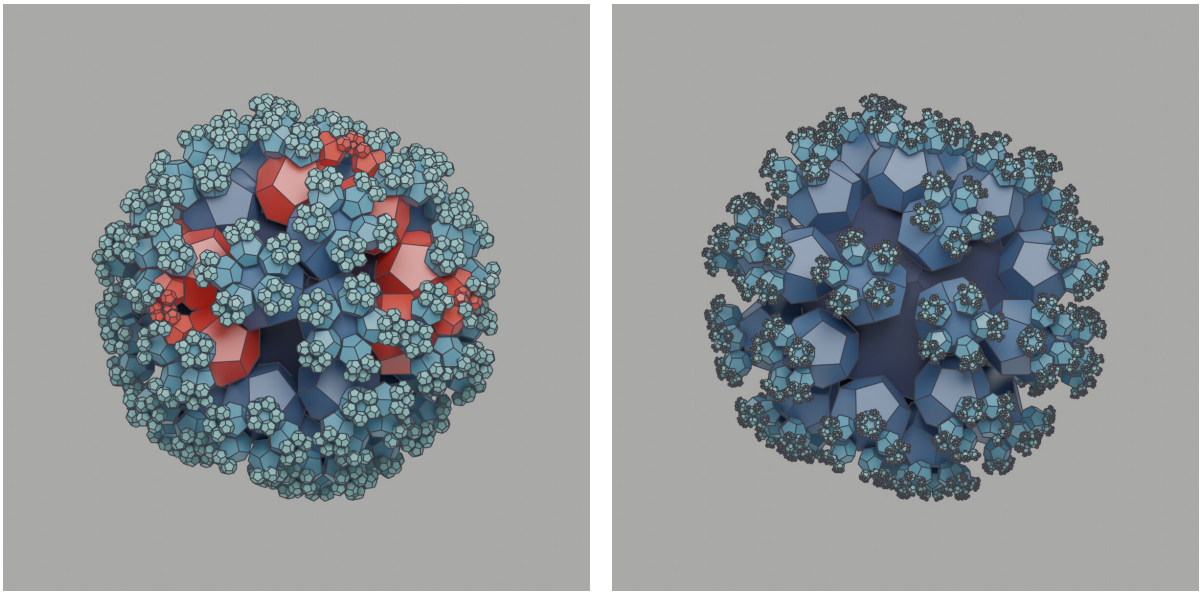


Figure 7: Growing from all twelve faces with optimal owners. Left: the artwork's proportions ($g \approx 0.435$, depth 3); the dodecahedra involved in clashes are red. Right: $g = 0.35$ (depth 4), certified self-avoiding.

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